Marmara University Engineering Faculty Electrical and Electronic Engineering Department EE 446.1/ EE 4046.1 Electrical Distribution

Name : Surname: Student Id:

Duration: 60 minutes 05/05/2023

Closed Book/Closed Notes. A simple calculator is allowed. No other electronics and no cell phones are permissible

Q1	10	10	10	**Q2**	5	5	5	10	**Q3**	15	15	15

I declare that I have not cheated, received, or given any assistance by any means throughout this exam. I am aware that violation of these principles according to the University regulations will result in a zero grade and a disciplinary investigation. I accept all rules and regulations regarding online exams. I permit the processing of my personal data as stated in the information text provided on the Faculty of Engineering website.

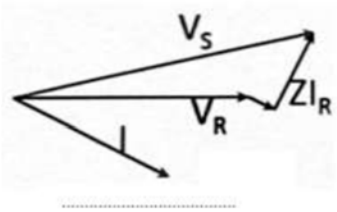
Question 1.)

a.)What are the power factor improvement equipment? (Write two factors)	b.) What are the advantages of radial circuits over networked circuits (Write 4 factors)	c.) What are the standard LV, MV, and HV levels in Turkiye;

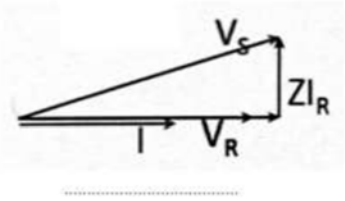
Question 2.) Fill in the blanks with proper wording

a.) The line constants of a medium transmission line are ……. distributed	b.) The length of a short transmission line is ……. mile	c.) The capacitance of a transmission line is …… element.

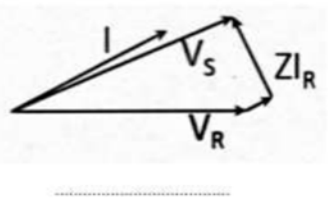
d.)Determine types of the phasor diagrams below;



(a)



(b)



(c)

Question3.) The underground cable line is supplying 8 kV to an 800 kVA 0.9 PF lagging single-phase load.

a.) What is the sending end voltage and current of this transmission line?
b.) What is the efficiency of the transmission line under these conditions?
c.) What is the voltage regulation of the transmission line under this condition?

Good luck

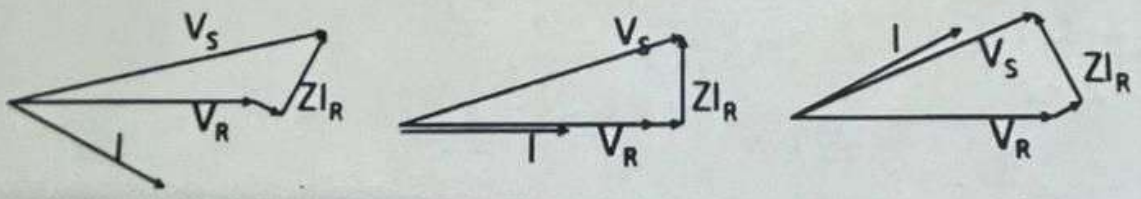
Question 1.)

a.)What is the power factor improvement equipment? (Write two factors)	b.) What are the advantages of radial circuits over networked circuits (Write 4 factors)	c.) What are the standard distribution and transmission voltage levels in Turkey;

Question 2.) Fill in the blanks with proper wording

a.) In short transmission lines, the effects of ..... are neglected.	b.) If the p.f. of the load decreases the line losses	

c.)Determine types of the phasor diagrams below;



Question3.) The underground cable line is supplying 8 kV to an 800 kVA 0.9 PF lagging single-phase load.(short transmission)

a.) What is the sending end voltage and current of this transmission line?
b.) What is the efficiency of the transmission line under these conditions?
c.) What is the voltage regulation of the transmission line under this condition?

Marmara University Engineering Faculty Electrical and Electronic Engineering Department EE 4046 Electrical Distribution Systems Final

Name : Surname: Student Id:

10/06/2024

Duration: 60 minutes
Closed Book/Closed Notes. A simple calculator is allowed. No other electronics and no cell phones are permissible

Q1	Q2	Q3
50	25	25

Question1.) A short transmission line that is terminated by a load of 200 kVA at a lagging Power factor of 0.866 at 2.4 kV. If the line impedance **Z** is 2.07+j0.661Ω,

calculate the sending end current,
the sending end voltage,
input impedance,
Real and reactive power loss in the line.

Question2.) Fill in the blanks with proper wording.

The Incoming _____ should have a _____ switch and a _____ _____. A disconnect switch is used to provide isolation since it cannot _____ the load current. A circuit breaker is used as a _____ device to interrupt fault currents automatically. Both _____ and _____ _____ may be operated locally or _____ from a supervisory control center

Question3.) What can trigger Blackouts in a Power Distribution System?

- overload
- equipment damage
- lightning / Storms
- short circuit / fault
- human errors

Good luck

Marmara University Engineering Faculty Electrical and Electronic Engineering
Department EE 4046 Electrical Distribution Systems Mid-term

Name : Surname: Student Id:
Duration: 60 minutes 07/04/2026
Closed Book/Closed Notes. A simple calculator is allowed. No other electronics and no cell phones are permissible

Q1	Q2	Q3	Q4	Q5
40	30	10	10	10

Q.1.) A load draws 100kW with a leading pf of 0.85. What are ø (power factor angle), Q and $|S|$? Draw the power triangle

Q.2.) There are three models based on the length of transmission lines

Q.3.) Write the following voltage levels for Türkiye.

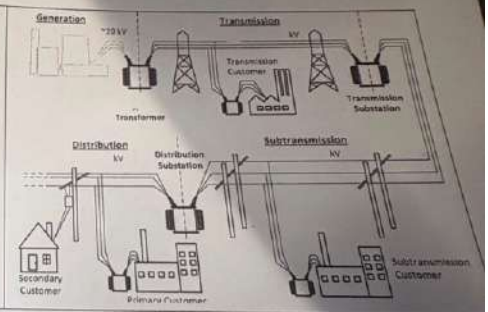
Generation ..20kV

transmission .380 kV

Sub transmission ..154 kV

distribution ...34.5 kV

and end-user ..400V



Q.4.) What is the main goal of distribution system planning?

4) Ana amaç, gelecekteki yük talebini:
güvenilir,
ekonomik,
güvenli,
kabul edilebilir gerilim sınırları içinde karşılamaktır.

Q.5.) Most common distribution system structure is ..... why?

Radial Distributions.
Nedenleri:
Basit yapılıdır.
Kurulum maliyeti düşüktür.
Koruma koordinasyonu kolaydır.
İşletmesi ve arıza tespiti kolaydır.

Q2
Short transmission line 80 km den az

medium trans. line 80km ile 250km arası

long tra. line 250 km den büyük